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General Education Reflection

My computer engineering classes are learning how to design and build a system, and I'm taking general education courses that will help me learn about the people and situations that the system will affect. So far, I have taken ECON 101, ECON 102, RUS 375, and INTST 235. These courses have afforded me a new perspective of engineering problems from an economic, cultural and global lens. I don't have all my U.S. Diversity requirements completed.

In ECON 101 I learned about scarcity, opportunity cost and supply and demand. These ideas relate to engineering in that all projects have finite resources and competing priorities. Engineers are faced with a dilemma when they have to select components, or design software, as they must also take into account cost as well as performance and reliability. Time and money spent in one feature means less time/money spent in another feature, a good consideration tool for these decisions. This becomes especially important for the digital divide, where a technologically successful product might not be used if individuals who need it are unable to pay for it.

I have learned through ECON 102 the broader context of economics: inflation, unemployment, economic growth etc. The ideas show the importance of taking factors outside the design into account when making engineering decisions. Costs can be affected by price

variation and resource availability, which affects project costs, manufacturing costs and feasibility of making a product. For example, if the semiconductors don't work, engineers have to consider other components. Engineers need to be aware of the economic context to make informed decisions that consider both technical needs and practical considerations.

In RUS 375 I was able to consider the perspectives of other cultures. But, when I studied another society, it really made me think twice about how people speak, how they think they understand information, how they think they understand information, how they act, what they do in what situation, you know. This is significant for engineering as people are unique and have expectations of what they are trying to use. Where it is easy for the developer to understand it could be difficult for other users. This is why it is important to take cultural differences into account to enable engineers to communicate effectively and create products that will benefit more clients.

In INTST 235 I had the opportunity to think of problems from a transnational point of view. Engineering projects frequently have people working from different countries, suppliers and users, so what decisions are made in one place can impact on another location. One of the sustainability of the environment is. Classical factors like functionality are not the only ones engineers have to take into account for designing electronic systems, they also have to take into account energy consumption, material consumption and electronic wastes. A global view makes me realize that a product can have ramifications beyond the use of the initial consumer.

The courses have made me think outside the box and have helped me with solving engineering problems in a different way. I must ask myself if a solution is affordable,

understandable and responsible and if it works. This will be a chance to revisit considerations for engineering based on varying experiences and needs as the U.S. Diversity requirement.